Mendelian randomization study of remnant cholesterol in relation to non-alcoholic fatty liver disease

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This document accompanies the main manuscript. It contains Supplemental Digital Content 1 (supplementary methods, run history and the STROBE-MR checklist) and Supplemental Digital Content 2 (supplementary tables S1–S12 and supplementary figures S1–S6). The complete tables, including all variant-level data, are in the Excel workbook Supplementary_Tables.xlsx (Supplemental Digital Content 3), and all code, console logs and intermediate files are in the public repository https://github.com/AliiReezaa/remnant-cholesterol-nafld-mr.

Table of contents. Overview.

Item	Content
SDC 1, Part A	Supplementary methods 1–8: data provenance, instrument selection flow, formulas, harmonisation, MR-PRESSO and extended estimator settings, validation, run history, software
SDC 1, Part B	STROBE-MR checklist with location in the manuscript
SDC 2, Tables	S1 instruments; S2 harmonised variant data; S3 all MR estimates; S4 heterogeneity and pleiotropy; S5 leave-one-out; S6 proxies; S7 positive control; S8 extended estimators; S9 heterogeneity (I^2, Egger, NOME); S10 influence diagnostics; S11 radial outliers; S12 cross-dataset concordance
SDC 2, Figures	S1 graphical abstract; S2–S3 diagnostics for Fairfield and FinnGen; S4 instrument strength, heterogeneity and influence; S5 radial MR; S6 positive-control diagnostics
SDC 3	Supplementary_Tables.xlsx

Supplemental Digital Content 1

Part A. Supplementary methods

1. Data sources and provenance. All data are public summary statistics. Table SM1 lists each source, accession and the details recorded when the data were extracted. The CAD file was downloaded on 15 October 2026 from the GWAS Catalog FTP in its genome build 37 harmonised form (239,499,711 bytes, 9,455,778 variants) and its MD5 checksum (0d296695219d8f4867bee625ac031ff0) matched the checksum published by EMBL-EBI. Case and control counts for the CAD meta-analysis (60,801 and 123,504) were taken from the study README on the GRASP site and not from the paper text, which could not be retrieved, and should be checked against the paper before submission.

Table SM1. Data sources used in the study.

Role	Source	Accession / identifier	Sample	Notes
Exposure: remnant cholesterol	Richardson et al, PLoS Biol 2022 (UK Biobank NMR, Nightingale)	GCST90092943 (OpenGWAS)	115,082, European ancestry	Inverse rank-normalised, SD units; BOLT-LMM
Outcome 1: NAFLD	Ghodsian et al, Cell Rep Med 2021	GCST90091033	8,434 cases / 770,180 controls	eMERGE, FinnGen freeze 4, UK Biobank, Estonian Biobank; EHR-defined
Outcome 2: NAFLD	Fairfield et al, Hepatol Commun 2022	GCST90054782	4,761 cases / 373,227 controls	UK Biobank; consensus diagnostic codes
Outcome 3: NAFLD	FinnGen release 5	finn-b-NAFLD (OpenGWAS)	894 cases / 217,898 controls	FinnGen
Positive control: CAD	Nikpay et al, Nat Genet 2015 (CARDIoGRAMplusC4D 1000G)	GCST003116	60,801 cases / 123,504 controls	Mixed ancestry, mainly European; build 37 harmonised file
LD reference	1000 Genomes Phase 3 European panel	–	503 individuals	Clumping ($r^2 < 0.001$, 10,000 kb) and proxies ($r^2 \geq 0.8$, 250 kb)

2. Instrument selection flow. 8,873 variants with $P < 5 \times 10^{-8}$ for remnant cholesterol (after removal of duplicated identifiers) → 7,593 present in the 1000 Genomes European panel (1,280 absent and not clumpable) → 52 independent variants after clumping ($r^2 < 0.001$, 10,000 kb, ranked by P). In the outcome datasets: Ghodsian, 3 variants missing without proxy and 6 variants replaced by proxies (Table S6), 1 palindromic variant removed, 48 used; Fairfield, 0 missing, 1 palindromic variant removed, 51 used; FinnGen, 3 missing without proxy, 3 proxies, 1 palindromic variant removed, 48 used. Positive control: 46 found by rsID, 2 by position, 2 by proxy, 2 absent, 1 palindromic removed, 49 used.

3. Formulas. Per-variant F statistic: $F = (\beta/SE)^2$. Variance explained by a variant, $R^2 = 2p(1-p)\beta^2$ with β in SD units; the overall F statistic was $F = R^2(N - k - 1) / ((1 - R^2)k)$, with R^2 the summed variance explained, k the number of variants and N the exposure sample size (159.2 for 48 variants, $R^2 = 6.23\%$, N = 115,082). Wald ratio: β_Y/β_X with standard error $SE_Y/|\beta_X|$. IVW estimate: a regression of β_Y on β_X weighted by $1/SE_Y^2$ with the intercept fixed at zero and multiplicative random-effects standard errors (residual standard error floored at one). Steiger test: for each variant the variance explained in the exposure was compared with that in the outcome; the latter was approximated on the latent logistic scale as $2p(1-p)\beta^2 / (2p(1-p)\beta^2 + \pi^2/3)$ without a prevalence assumption. Minimum detectable effect with 80% power: $|\log OR| = (z_{0.975} + z_{0.80}) \times SE_{IVW}$, i.e. $2.80 \times SE_{IVW}$. Bias from complete sample overlap relative to the confounded association: approximately $1/F_{overall}$.

4. Harmonisation. Effects were aligned to the RC-increasing allele. Proxies were selected as the variant with the highest $r^2 \geq 0.8$ within 250 kb in the 1000 Genomes European panel and alleles were aligned by LD phase. Palindromic variants with effect allele frequency between 0.42 and 0.58 were excluded; other palindromic variants were aligned by allele frequency. No allele mismatches were found.

5. MR-PRESSO and extended estimator settings. MR-PRESSO was run with the MRPRESSO R package with 10,000 simulations, outlier and distortion tests, a significance threshold of 0.05 for outlier detection and seed 2026; the raw and outlier-corrected estimates, the global test and the outlier list are in the repository (results/mrpresso_results.csv). Extended estimators were run with the MendelianRandomization package (maximum likelihood, MR-Lasso, contamination mixture, debiased IVW, IVW fixed and random effects) with default settings, with mr.raps (Tukey and Huber losses with overdispersion) and with RadialMR (IVW radial, modified second-order weights; outliers flagged at the Bonferroni-adjusted 5% level of the individual Q statistic). The random seed was 2026. Influence diagnostics used Cook's distance (threshold 4/n) and studentized residuals from the weighted regression of β_Y on β_X through the origin.

6. Validation of the Python implementation. Before use, the Python code was compared with R output of a previous analysis of these data: IVW and MR-Egger estimates and standard errors, the Egger intercept, Cochran's Q and the MR-PRESSO global statistic (RSSobs 186.53 in both) and outlier-corrected estimate were reproduced. For the positive control, the Python and R (TwoSampleMR) analyses gave identical IVW and MR-Egger estimates and the same weighted median to the third decimal; the weighted mode point estimate (1.408 in Python, 1.393 in R) and the bootstrap standard errors of the simple mode differ slightly, and the reason for the difference in the weighted mode was not investigated further (Table S7).

7. Run history. Table SM2 lists every analysis run, including runs that were superseded, so that the reported results can be traced.

Table SM2. Run history of the analyses.

Run	Analysis	Software	Outcome / note
1	Primary NAFLD analysis, 52-variant set, three datasets	Python (custom TwoSampleMR-equivalent code)	Results reported in main text
2	MR-PRESSO confirmation	R MRPRESSO 1.0, 10,000 simulations, seed 2026	Same outliers as Python; FinnGen corrected P .250 (R) vs .24 (Python); R values reported
3	Sensitivity sets: APOE region excluded; earlier 48-variant set	Python	Reported in Table 4
4	Positive control (CAD), Python	Python	Reported (Table S7)
5	Positive control, R run 1 (first run)	R 4.4.3, TwoSampleMR 0.6.22	Superseded: logging only; kept in

			repository
6	Positive control, R run 2 (logged, unseeded)	R 4.4.3	Superseded: bootstrap standard errors for mode estimators varied between runs; kept in repository
7	Positive control, R run 3 (final, seed 2026)	R 4.4.3	Reported in Table 6
8	Extended estimators, influence, radial, I^2 (three datasets)	R 4.4.3; MendelianRandomization 0.10.0, mr.raps 0.2, RadialMR 1.2.1	Reported in Table 5 and Tables S8–S11
9	Manuscript figures and concordance statistics	Python 3 (matplotlib, scipy)	Script code/s13_manuscript_figures.py

8. Software. Python 3 (numpy, pandas, scipy, matplotlib); R 4.4.3 (2025-02-28) with TwoSampleMR 0.6.22, MRPRESSO 1.0, MendelianRandomization 0.10.0, mr.raps 0.2 and RadialMR 1.2.1. The study was not pre-registered.

Part B. STROBE-MR checklist

Table SM3. STROBE-MR checklist (item wording paraphrased).

Item	Recommendation	Location
1	Identify MR as the design in the title or abstract	Title; Abstract
2	Background and rationale, including causal plausibility and why MR is suitable	1. Introduction
3	Specific objectives and causal hypotheses	1. Introduction (final paragraph)
4a	Setting, design, underlying population, dates	2.1; Table 1
4b	Participants, eligibility, sample size, power calculations	2.1; 2.7; Table 1
4c	Measurement, quality control and selection of genetic variants	2.2
4d	Methods and diagnostic criteria for exposure and outcomes	2.1 (NMR remnant cholesterol; NAFLD case definitions of the source GWAS)
4e	Ethics approval and consent	Information block; Acknowledgments (public summary data; no new approval)
5	State the three IV assumptions	2.1
6a–c	Quantitative variables, genetic variants and weights, estimator and covariates	2.1–2.3
6d	Handling of missing data	2.2 (proxies, dropped variants)
6e	Multiple testing	2.3 (no adjustment; one relation in three datasets)
7	Methods to assess the IV assumptions	2.2 (F, Steiger); 2.4; 2.8
8	Sensitivity and additional analyses	2.3–2.8
9a	Software, packages, versions, settings	2.9; SDC 1 Part A (8)
9b	Pre-registration	2.1 (not pre-registered)
10a	Numbers at each study stage and reasons for exclusion	2.2; Fig. 1; SDC 1 Part A (2)
10b	Summary statistics for exposure and outcomes	Table 1
10c	Heterogeneity across meta-analysed studies	Not applicable (summary statistics used as published)
10d	Similarity of variant–exposure associations across samples; overlapping individuals	Overlap: 2.7, 3.6. Similarity of exposure associations across samples was not tested separately (all samples European ancestry); stated as a limitation
11a	Variant–exposure and variant–outcome associations	Fig. 3, 4; Tables S1, S2
11b	MR estimates with uncertainty on an interpretable scale	3.1; Table 2; Fig. 2
11c	Absolute risk translation	Not done (null result)
11d	Plots (forest, scatter)	Fig. 2, 4, 5; Figs. S2, S3
12a, b	Assessment of assumptions; I^2, Q	3.3; Table 3; Tables S4, S9
13a, b	Robustness to assumption violations; other sensitivity analyses	3.4, 3.5; Tables 4, 5
13c	Assessment of causal direction (bidirectional MR)	Not done; Steiger filter used; stated as a limitation
13d	Comparison with non-MR estimates	4. Discussion (published observational studies)
13e	Additional plots (leave-one-out)	Fig. 5C; Figs. S2, S3
14	Key results with reference to objectives	4. Discussion (first paragraph)
15	Limitations: IV assumptions, bias, imprecision	4. Discussion (limitations paragraph)
16a–c	Cautious interpretation, mechanisms, clinical relevance	4. Discussion; 5. Conclusion
17	Generalizability	4. Discussion (seventh limitation)
18	Funding	Information block
19	Data and code sharing	Data and code availability
20	Conflicts of interest	Information block

Supplemental Digital Content 2

Supplementary tables

Table S1. The 52 remnant cholesterol instruments.

SNP	Chr	Position (build 37)	EA/OA	EAF	β (SD)	SE	P	F	R^2 (%)	Used*
rs72654473	19	45,414,399	C/A	0.892	0.264	0.007	< 1e-300	1598	1.37	GFN
rs12151108	19	11,197,261	G/A	0.880	0.153	0.006	7.0e-130	588	0.51	GFN
rs157580	19	45,395,266	G/A	0.395	-0.084	0.004	5.1e-89	400	0.35	GFN
rs11591147	1	55,505,647	G/T	0.983	0.309	0.016	5.6e-87	391	0.34	GFN
rs12740374	1	109,817,590	G/T	0.779	0.096	0.005	1.4e-84	380	0.33	GFN
rs693	2	21,232,195	G/A	0.479	-0.077	0.004	1.6e-78	352	0.30	GFN
rs261290	15	58,678,720	T/C	0.345	0.076	0.004	4.1e-69	309	0.27	GFN
rs112875651	8	126,506,694	G/A	0.608	0.069	0.004	1.7e-58	260	0.23	GFN
rs58542926	19	19,379,549	C/T	0.926	0.118	0.008	1.2e-51	229	0.20	GFN
rs247616	16	56,989,590	C/T	0.675	0.061	0.004	7.4e-44	193	0.17	GFN
rs12916	5	74,656,539	T/C	0.600	-0.058	0.004	8.7e-44	193	0.17	GFN
rs7534572	1	62,999,675	C/G	0.353	-0.056	0.004	9.4e-39	170	0.15	–F–
rs633695	15	58,725,839	A/G	0.708	-0.055	0.005	1.4e-33	146	0.13	GFN
rs964184	11	116,648,917	G/C	0.133	0.071	0.006	6.6e-32	138	0.12	GFN
rs115478735	9	136,149,711	A/T	0.817	-0.059	0.005	3.8e-29	126	0.11	GFN
rs77542162	17	67,081,278	A/G	0.977	-0.149	0.014	2.8e-27	117	0.10	GFN
rs4299376	2	44,072,576	G/T	0.324	0.047	0.004	7.1e-27	115	0.10	GFN
rs102275	11	61,557,803	T/C	0.650	0.045	0.004	3.9e-26	112	0.10	GFN
rs4704834	5	156,443,066	A/G	0.356	-0.041	0.004	5.1e-22	93	0.08	GFN
rs574454081	19	45,305,530	G/A	0.801	-0.062	0.007	3.2e-21	89	0.08	–F–
rs72848251	6	32,601,288	G/A	0.805	-0.048	0.005	7.5e-20	83	0.07	GFN
rs1883711	20	39,179,822	G/C	0.969	-0.109	0.012	2.1e-19	81	0.07	GFN
rs4665972	2	27,598,097	T/C	0.395	0.036	0.004	1.6e-17	73	0.06	GFN
rs3777411	6	160,476,945	C/T	0.849	0.048	0.006	3.2e-17	71	0.06	GFN
rs11206517	1	55,526,428	T/G	0.967	-0.091	0.011	2.5e-15	63	0.05	GFN
rs2738447	19	11,227,480	A/C	0.407	-0.033	0.004	3.6e-15	62	0.05	GFN
rs34042070	16	72,101,525	C/G	0.813	-0.041	0.005	6.2e-15	61	0.05	GFN
rs926053	19	44,952,075	G/C	0.889	0.050	0.007	1.4e-14	59	0.05	GFN
rs77960347	18	47,109,955	A/G	0.987	-0.135	0.018	5.7e-14	56	0.05	GFN
rs56325564	17	45,766,771	G/A	0.517	-0.030	0.004	2.1e-13	54	0.05	GFN
rs472495	1	55,521,313	G/T	0.351	-0.031	0.004	2.5e-13	54	0.05	GFN
rs12804228	11	126,260,684	A/G	0.915	-0.052	0.007	1.9e-12	50	0.04	GFN
rs72631343	17	67,191,270	C/G	0.871	0.043	0.006	2.2e-12	49	0.04	GFN
rs553427	1	234,852,760	C/T	0.483	-0.029	0.004	2.6e-12	49	0.04	GFN
rs11854242	15	58,571,910	C/T	0.725	0.032	0.005	6.2e-12	47	0.04	GFN
rs1169292	12	121,426,478	C/T	0.690	-0.030	0.004	7.6e-12	47	0.04	GFN
rs838146	19	49,248,052	C/T	0.500	0.028	0.004	1.3e-11	46	0.04	GFN
rs6601299	8	9,184,691	T/C	0.101	-0.045	0.007	4.5e-11	43	0.04	GFN
rs10260606	7	44,584,551	G/C	0.817	-0.034	0.005	1.8e-10	41	0.04	GFN
rs2326077	8	59,385,919	C/T	0.337	0.027	0.004	4.7e-10	39	0.03	GFN
rs2737245	8	116,658,583	G/T	0.721	0.028	0.005	1.1e-09	37	0.03	GFN
rs653178	12	112,007,756	C/T	0.484	-0.025	0.004	1.6e-09	36	0.03	GFN
rs72843670	6	29,812,895	T/C	0.862	-0.035	0.006	2.6e-09	35	0.03	GFN
rs532754706	6	161,329,915	A/G	0.977	0.096	0.016	3.3e-09	35	0.03	–F–
rs9616847	22	50,868,669	A/T	0.612	-0.025	0.004	3.3e-09	35	0.03	GFN
rs4939883	18	47,167,214	T/C	0.181	-0.031	0.005	5.9e-09	34	0.03	GFN
rs1500188	5	122,827,953	A/G	0.550	0.024	0.004	6.0e-09	34	0.03	GFN
rs35633876	2	20,363,074	G/T	0.518	0.024	0.004	6.9e-09	34	0.03	GFN
rs1293261	1	25,713,234	no/no	–	–	–	8.4e-09	–	–	——
rs6602911	13	114,547,372	C/T	0.640	-0.025	0.004	9.0e-09	33	0.03	GFN
rs72694391	14	24,874,026	T/C	0.518	-0.023	0.004	3.6e-08	30	0.03	GFN
rs12620918	2	63,268,355	T/C	0.826	-0.030	0.005	4.8e-08	30	0.03	GFN

*G = Ghodsian, F = Fairfield, N = FinnGen; – = not used (missing without proxy or palindromic). EA = effect (RC-raising) allele; OA = other allele; EAF = effect allele frequency. Variant-level outcome data are in Supplementary_Tables.xlsx (S2).

S2. Harmonised variant-level data (exposure and outcome effect, standard error, P value, Wald ratio, Steiger direction) for the three NAFLD datasets and the positive control are in Supplementary_Tables.xlsx (sheets S2 and S7) and in the repository folder analysis/primary_clumped52.

Table S3. All MR estimates for the primary 52-variant set (Python), by analysis set and method.

Dataset	Analysis set	Method	SNPs	OR	95% CI	P

Ghodsian	all SNPs	IVW	48	1.045	0.864–1.265	.650
Ghodsian	all SNPs	MR-Egger	48	0.901	0.654–1.240	.524
Ghodsian	all SNPs	Weighted median	48	1.108	0.954–1.287	.178
Ghodsian	all SNPs	Simple mode	48	1.006	0.791–1.280	.960
Ghodsian	all SNPs	Weighted mode	48	1.071	0.936–1.226	.321
Ghodsian	Steiger-filtered	IVW	45	1.081	0.943–1.238	.263
Ghodsian	Steiger-filtered	MR-Egger	45	1.066	0.847–1.342	.589
Ghodsian	Steiger-filtered	Weighted median	45	1.108	0.958–1.282	.166
Ghodsian	Steiger-filtered	Simple mode	45	1.013	0.794–1.292	.918
Ghodsian	Steiger-filtered	Weighted mode	45	1.073	0.928–1.242	.346
Ghodsian	MR-PRESSO outliers removed	IVW	44	1.023	0.914–1.145	.693
Ghodsian	MR-PRESSO outliers removed	MR-Egger	44	1.020	0.845–1.232	.835
Ghodsian	MR-PRESSO outliers removed	Weighted median	44	1.108	0.959–1.280	.165
Ghodsian	MR-PRESSO outliers removed	Simple mode	44	1.013	0.785–1.308	.922
Ghodsian	MR-PRESSO outliers removed	Weighted mode	44	1.073	0.931–1.237	.335
Fairfield	all SNPs	IVW	51	0.999	0.776–1.286	.994
Fairfield	all SNPs	MR-Egger	51	0.754	0.492–1.155	.201
Fairfield	all SNPs	Weighted median	51	1.022	0.832–1.256	.833
Fairfield	all SNPs	Simple mode	51	0.792	0.565–1.110	.182
Fairfield	all SNPs	Weighted mode	51	0.983	0.803–1.203	.867
Fairfield	Steiger-filtered	IVW	47	0.996	0.865–1.146	.951
Fairfield	Steiger-filtered	MR-Egger	47	0.923	0.727–1.172	.513
Fairfield	Steiger-filtered	Weighted median	47	1.021	0.844–1.236	.828
Fairfield	Steiger-filtered	Simple mode	47	0.767	0.528–1.116	.172
Fairfield	Steiger-filtered	Weighted mode	47	1.002	0.819–1.225	.986
Fairfield	MR-PRESSO outliers removed	IVW	48	1.009	0.871–1.169	.907
Fairfield	MR-PRESSO outliers removed	MR-Egger	48	0.897	0.700–1.150	.395
Fairfield	MR-PRESSO outliers removed	Weighted median	48	1.022	0.847–1.233	.823
Fairfield	MR-PRESSO outliers removed	Simple mode	48	0.780	0.536–1.135	.201
Fairfield	MR-PRESSO outliers removed	Weighted mode	48	0.989	0.804–1.217	.921
FinnGen	all SNPs	IVW	48	0.848	0.561–1.281	.433
FinnGen	all SNPs	MR-Egger	48	0.863	0.426–1.745	.683
FinnGen	all SNPs	Weighted median	48	1.082	0.700–1.673	.722
FinnGen	all SNPs	Simple mode	48	0.839	0.387–1.816	.658
FinnGen	all SNPs	Weighted mode	48	0.900	0.598–1.355	.617
FinnGen	Steiger-filtered	IVW	35	0.949	0.706–1.275	.727
FinnGen	Steiger-filtered	MR-Egger	35	0.918	0.562–1.502	.737
FinnGen	Steiger-filtered	Weighted median	35	1.134	0.704–1.827	.605
FinnGen	Steiger-filtered	Simple mode	35	0.894	0.423–1.889	.771
FinnGen	Steiger-filtered	Weighted mode	35	1.017	0.613–1.688	.948
FinnGen	MR-PRESSO outliers removed	IVW	46	0.839	0.624–1.127	.244
FinnGen	MR-PRESSO outliers removed	MR-Egger	46	0.952	0.580–1.564	.848
FinnGen	MR-PRESSO outliers removed	Weighted median	46	1.039	0.682–1.582	.858
FinnGen	MR-PRESSO outliers removed	Simple mode	46	0.843	0.428–1.661	.623
FinnGen	MR-PRESSO outliers removed	Weighted mode	46	0.909	0.572–1.444	.688

The MR-PRESSO-corrected IVW estimates reported in the main text (R package) are in Table 4. Estimates for the APOE-excluded and earlier 48-variant sets are in Supplementary_Tables.xlsx and the repository.

Table S4. Heterogeneity, pleiotropy and Steiger filtering.

Dataset	SNPs	Q IVW	P	Q Egger	P	Egger intercept (SE)	P	Steiger-filtered IVW OR	Removed
Ghodsian	48	204.8	1.4e-21	199.2	5.8e-21	0.0112 (0.0099)	.261	1.081	3
Fairfield	51	251.4	1.2e-28	239.0	8.1e-27	0.0211 (0.0133)	.118	0.996	4
FinnGen	48	106.2	1.8e-06	106.2	1.2e-06	-0.0013 (0.0215)	.952	0.949	13

Table S5. Leave-one-out summary for the IVW estimate.

Dataset	Runs	OR range	Smallest P	Three most influential omitted variants (OR when omitted)
Ghodsian	48	0.991–1.121	0.150	rs58542926 (1.121), rs112875651 (0.991), rs157580 (1.073)
Fairfield	51	0.932–1.123	0.261	rs58542926 (1.123), rs112875651 (0.932), rs72654473 (0.958)
FinnGen	48	0.735–0.966	0.102	rs112875651 (0.735), rs58542926 (0.966), rs72654473 (0.792)

Every single-omission estimate is in Supplementary_Tables.xlsx (S5).

Table S6. Proxy variants used in the NAFLD outcome datasets.

Outcome	Target variant	Proxy	r^2	Distance (bp)
Ghodsian	rs247616	rs247617	1.000	1,126
Ghodsian	rs115478735	rs635634	0.981	5,289
Ghodsian	rs72848251	rs35062987	0.801	10,553
Ghodsian	rs56325564	rs3809868	0.922	16,175
Ghodsian	rs12804228	rs35823964	0.931	3,288
Ghodsian	rs838146	rs838147	1.000	1,186
FinnGen	rs247616	rs247617	1.000	1,126
FinnGen	rs72848251	rs17205170	0.862	1,195
FinnGen	rs12804228	rs35823964	0.931	3,288

Three variants (rs7534572, rs574454081, rs532754706) had no proxy and were dropped from the Ghodsian and FinnGen analyses.

Table S7. Positive control (CAD): Python and R estimates side by side (49 variants).

Method	Python OR (95% CI)	P	R OR (95% CI)	P
MR Egger	1.871 (1.461–2.395)	9.4e-06	1.871 (1.461–2.395)	9.4e-06
Weighted median	1.475 (1.321–1.648)	5.9e-12	1.475 (1.321–1.648)	5.6e-12
Inverse variance weighted	1.579 (1.378–1.809)	5.2e-11	1.579 (1.378–1.809)	5.2e-11
Simple mode	1.408 (1.072–1.849)	1.7e-02	1.408 (1.088–1.821)	1.2e-02
Weighted mode	1.408 (1.241–1.597)	2.8e-06	1.393 (1.216–1.597)	1.7e-05

Cochran's Q 276.9 (48 df), P = 6.2e-34; Egger intercept -0.0115 (SE 0.0072), P = 0.116. MR-PRESSO (R): global P < .0001; 6 outliers (rs115478735, rs12151108, rs12740374, rs1500188, rs3777411, rs653178); corrected OR 1.492 (1.348–1.651); distortion test P = .175. Python: 5 outliers, corrected OR 1.540. Steiger removed rs653178 (IVW OR 1.602). Leave-one-out IVW OR 1.51–1.62, all $P < 10^{-8}$. Substitutions for missing instruments: rs115478735 → rs507666 (r^2 0.988); rs838146 → rs838147 (r^2 1.0); rs72848251 → rs115129597 (same position); rs72843670 → rs115366102 (same position); rs574454081 and rs532754706 dropped. R run 1 and run 2 are kept in the repository.

Table S8. Extended estimators for all datasets.

Dataset	Estimator	SNPs	OR	95% CI	P	Note
Ghodsian	Maximum likelihood	48	1.047	0.862–1.270	.645	
Ghodsian	MR-Lasso	48	1.040	0.942–1.149	.438	valid instruments: 40
Ghodsian	Contamination mixture	48	1.062	0.942–1.197	–	CI 0.932-1.185; valid 33
Ghodsian	Debiased IVW	48	1.046	0.873–1.253	.629	
Ghodsian	IVW (random effects, MendelianRandomization)	48	1.045	0.864–1.265	.650	
Ghodsian	IVW (fixed effect)	48	1.045	0.954–1.145	.343	
Ghodsian	MR-RAPS (tukey, overdispersed)	48	1.033	0.925–1.153	.565	
Ghodsian	MR-RAPS (huber, overdispersed)	48	1.022	0.883–1.183	.772	
Ghodsian	IVW, variants with F>=30	47	1.046	0.862–1.269	.648	
Ghodsian	IVW, variants with MAF>=0.01	48	1.045	0.864–1.265	.650	
Ghodsian	Radial IVW (modified second order weights)	48	1.045	0.864–1.265	.650	outliers: rs157580;rs112875651;rs58542926;rs4665972;rs56325564;rs553427;rs2326077;rs2737245
Fairfield	Maximum likelihood	51	0.999	0.773–1.292	.994	
Fairfield	MR-Lasso	51	1.016	0.893–1.156	.809	valid instruments: 45
Fairfield	Contamination mixture	51	0.994	0.802–1.232	–	CI 0.766-1.178; valid 41
Fairfield	Debiased IVW	51	0.999	0.786–1.270	.994	
Fairfield	IVW (random effects, MendelianRandomization)	51	0.999	0.776–1.286	.994	
Fairfield	IVW (fixed effect)	51	0.999	0.893–1.118	.987	
Fairfield	MR-RAPS (tukey, overdispersed)	51	1.010	0.882–1.158	.883	
Fairfield	MR-RAPS (huber, overdispersed)	51	0.991	0.798–1.232	.938	
Fairfield	IVW, variants with F>=30	50	0.993	0.770–1.280	.958	
Fairfield	IVW, variants with MAF>=0.01	51	0.999	0.776–1.286	.994	
Fairfield	Radial IVW (modified second order weights)	51	0.999	0.776–1.286	.994	outliers: rs157580;rs112875651;rs58542926;rs1883711;rs4665972;rs1169292;rs2326077
FinnGen	Maximum likelihood	48	0.846	0.559–1.282	.431	
FinnGen	MR-Lasso	48	0.839	0.624–1.127	.244	valid instruments: 46
FinnGen	Contamination mixture	48	1.063	0.705–1.601	–	CI 0.698-1.585; valid 38

FinnGen	Debiased IVW	48	0.847	0.572–1.253	.406	
FinnGen	IVW (random effects, MendelianRandomization)	48	0.848	0.561–1.281	.433	
FinnGen	IVW (fixed effect)	48	0.848	0.644–1.116	.238	
FinnGen	MR-RAPS (tukey, overdispersed)	48	0.868	0.644–1.170	.353	
FinnGen	MR-RAPS (huber, overdispersed)	48	0.856	0.591–1.239	.410	
FinnGen	IVW, variants with F>=30	47	0.861	0.570–1.301	.478	
FinnGen	IVW, variants with MAF>=0.01	48	0.848	0.561–1.281	.433	
FinnGen	Radial IVW (modified second order weights)	48	0.848	0.561–1.281	.433	outliers: rs112875651;rs58542926;rs6601299

The contamination mixture model does not provide a P value; its interval is a likelihood-based confidence interval. MR-Lasso and the contamination mixture model also report the number of variants judged valid.

Table S9. Heterogeneity, I^2 and no-measurement-error (NOME) statistics.

Dataset	SNPs	Q IVW	I^2 (%)	Q Egger	I^2_{GX} (%)	Egger intercept (SE)	P	φ
Ghodsian	48	204.8	77.1	199.2	99.35	0.0112 (0.0099)	.255	4.36
Fairfield	51	251.4	80.1	239.0	99.33	0.0211 (0.0133)	.112	5.03
FinnGen	48	106.2	55.7	106.2	99.35	-0.0013 (0.0215)	.952	2.26

φ = multiplicative random-effects residual standard error (values above 1 indicate overdispersion). These values come from the R run (MendelianRandomization package), which tests the Egger intercept against a normal distribution; Table 3 gives the t-distribution P values from the primary analysis (.261, .118 and .952). The conclusion is the same.

Table S10. Variants exceeding the Cook's distance threshold (4/n).

Dataset	SNP	Cook's distance	Studentized residual	Leverage	Threshold
Fairfield	rs58542926	0.820	-5.71	0.039	0.078
Fairfield	rs112875651	0.288	3.11	0.034	0.078
Fairfield	rs72654473	0.107	0.65	0.202	0.078
Fairfield	rs157580	0.089	1.28	0.052	0.078
Fairfield	rs4665972	0.081	3.11	0.010	0.078
FinnGen	rs112875651	0.460	3.89	0.038	0.083
FinnGen	rs58542926	0.386	-4.54	0.026	0.083
FinnGen	rs72654473	0.103	0.72	0.163	0.083
Ghodsian	rs58542926	0.521	-5.14	0.030	0.083
Ghodsian	rs112875651	0.302	3.04	0.037	0.083

Full diagnostics for all variants are in Supplementary_Tables.xlsx.

Table S11. Radial MR outliers.

Dataset	Outliers	Variants
Ghodsian	8	rs157580, rs112875651, rs58542926, rs4665972, rs56325564, rs553427, rs2326077, rs2737245
Fairfield	7	rs157580, rs112875651, rs58542926, rs1883711, rs4665972, rs1169292, rs2326077
FinnGen	3	rs112875651, rs58542926, rs6601299

Table S12. Cross-dataset concordance of single-variant Wald ratios.

Pair	Variants	Spearman ρ	P	Same sign (%)
Ghodsian 2021 vs Fairfield 2022	48	0.678	1.2×10^{-7}	66.7
Ghodsian 2021 vs FinnGen R5	48	0.487	4.5×10^{-4}	62.5
Fairfield 2022 vs FinnGen R5	48	0.328	.023	62.5

Datasets share participants, so concordance is expected to exceed that of independent replications.

Supplementary figures

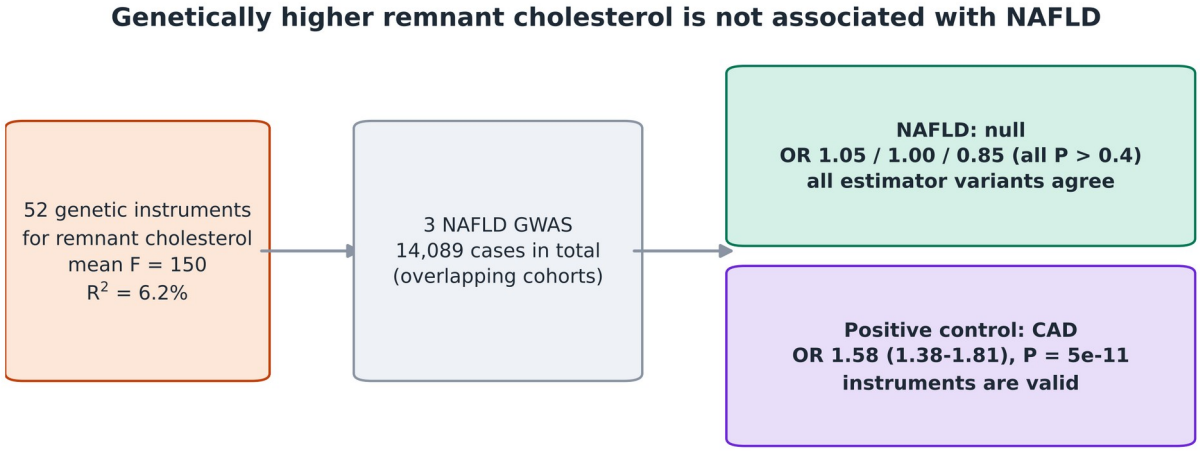


Figure S1. Graphical abstract. Fifty-two genetic instruments for remnant cholesterol were tested against three NAFLD datasets (null) and against coronary artery disease as a positive control (strong association).

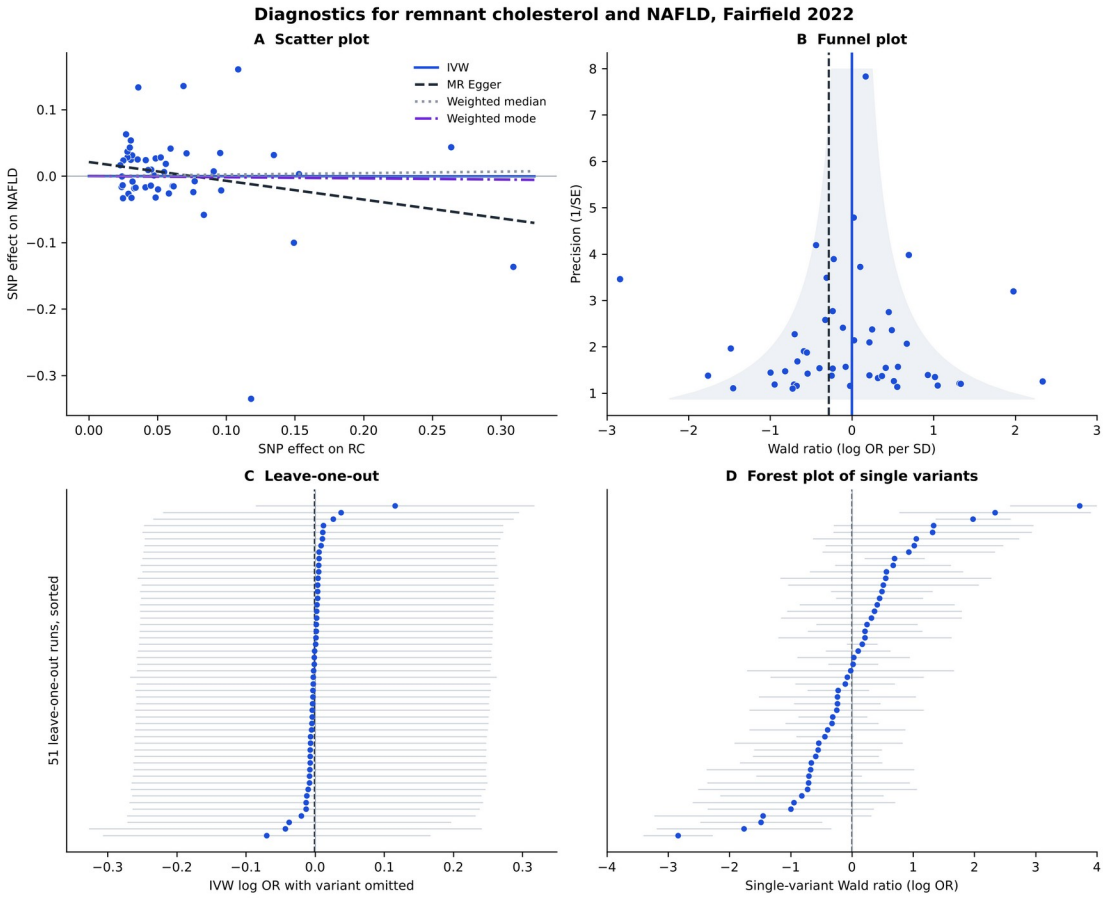


Figure S2. Diagnostics for remnant cholesterol and NAFLD in the Fairfield (UK Biobank) dataset. Panel A: scatter plot with IVW, MR-Egger, weighted median and weighted mode fits; panel B: funnel plot; panel C: leave-one-out; panel D: single-variant forest plot.

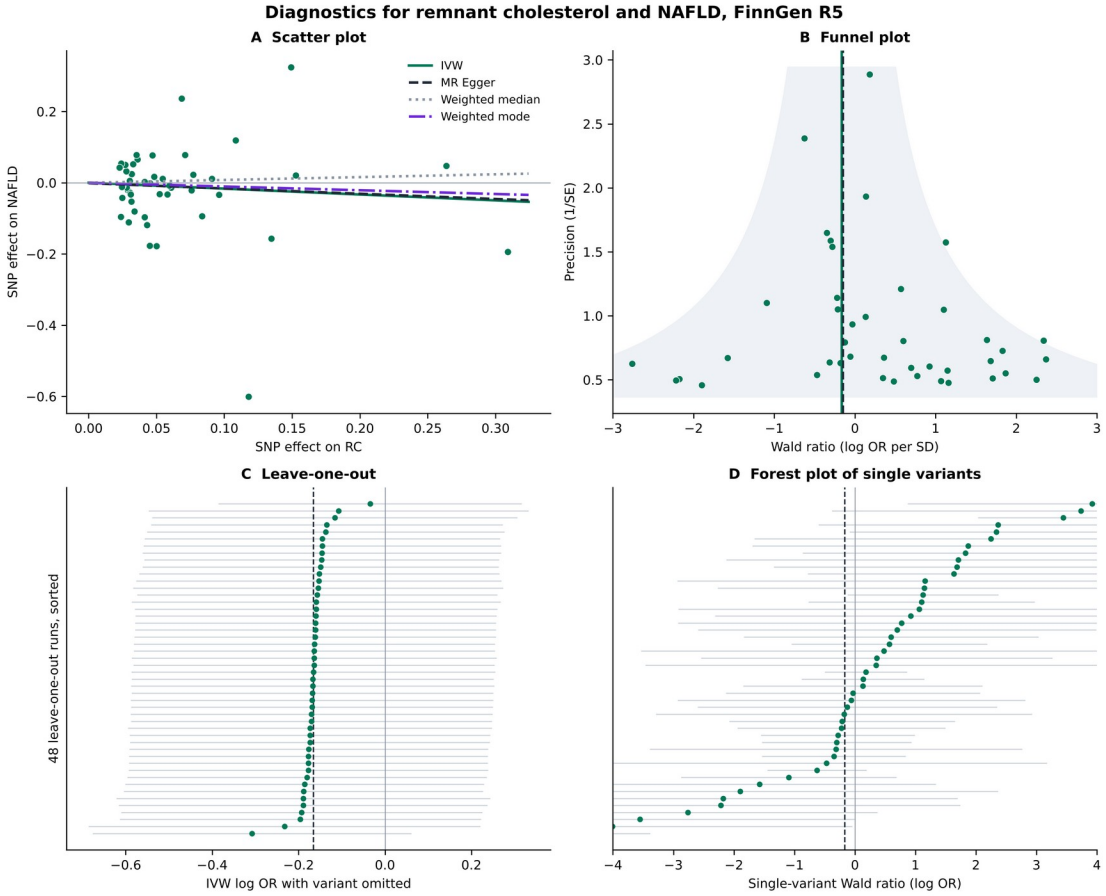


Figure S3. Diagnostics for remnant cholesterol and NAFLD in the FinnGen release 5 dataset. Panels as in Fig. S2. The wide intervals reflect the small number of cases (894).

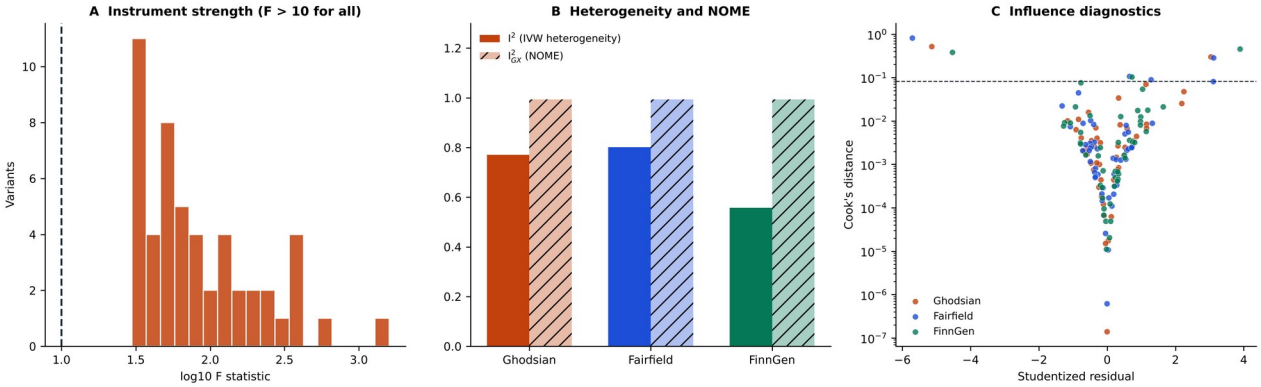


Figure S4. Instrument strength, heterogeneity and influence. Panel A: distribution of per-variant F statistics (dashed line: F = 10). Panel B: I^2 of the IVW estimate and I^2_{GX} of the MR-Egger framework in each dataset. Panel C: studentized residuals against Cook's distance (dashed line: 4/n); the points above the line are rs58542926 (TM6SF2) and rs112875651 (TRIB1) in all datasets.

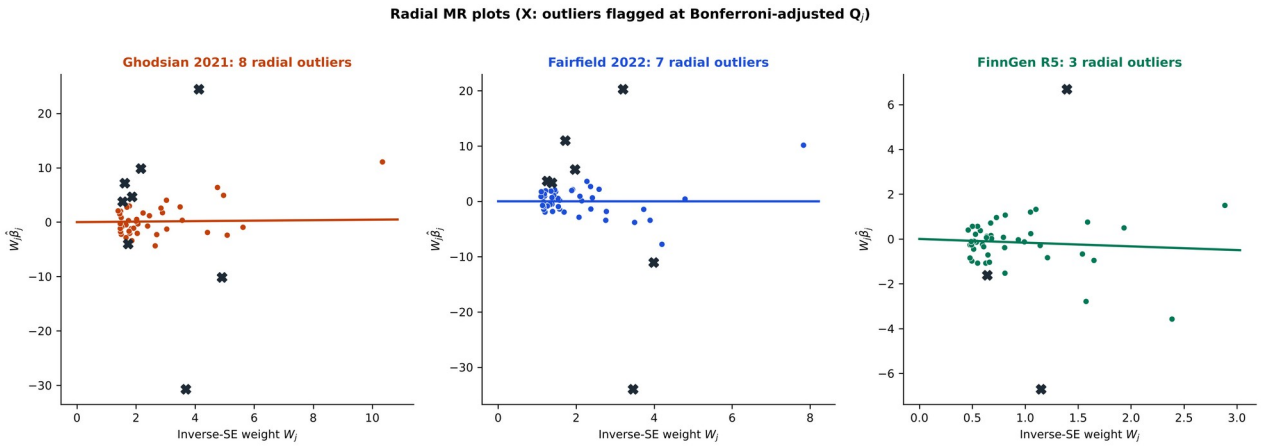


Figure S5. Radial MR plots. Each point is a variant plotted by inverse-standard-error weight and weighted ratio estimate; the line is the radial IVW estimate and crosses mark variants flagged as outliers.

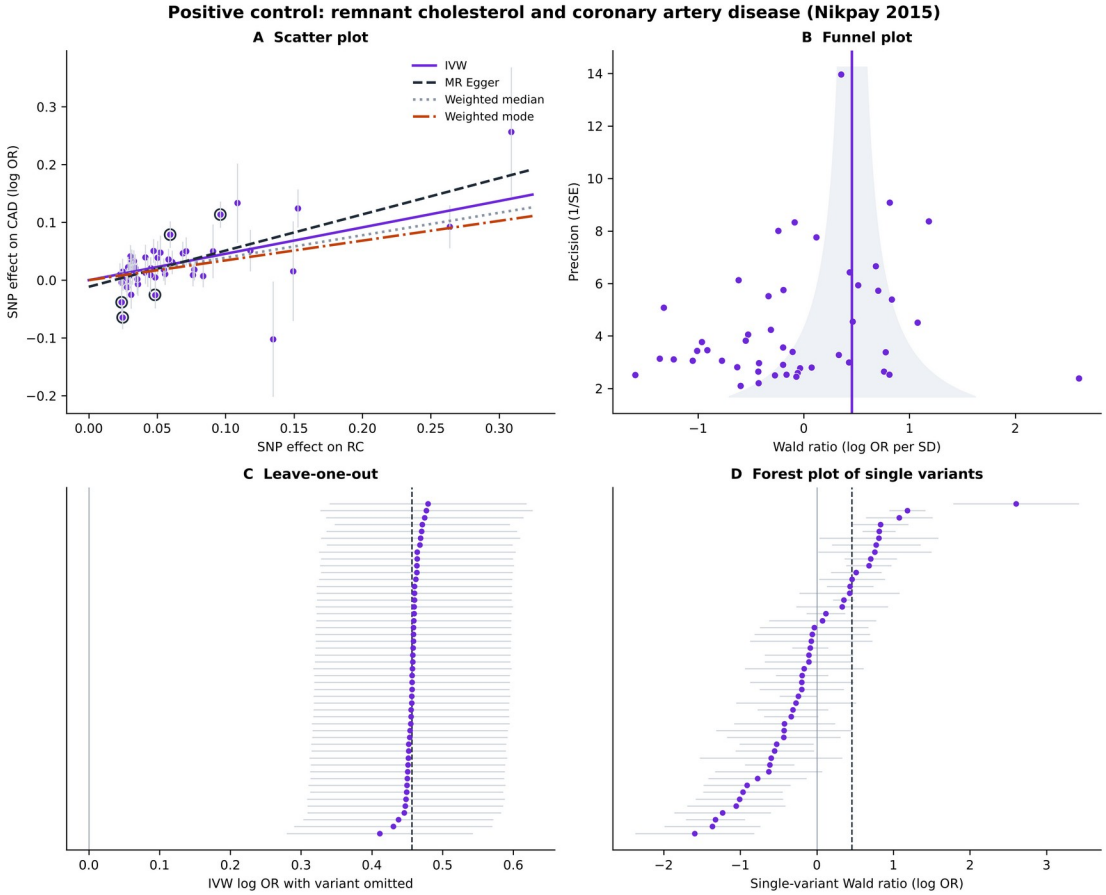


Figure S6. Diagnostics for the positive control (remnant cholesterol and coronary artery disease). Panel A: scatter plot (circled points are MR-PRESSO outliers in the Python analysis); panel B: funnel plot; panel C: leave-one-out analysis, in which no single omission removes the association; panel D: single-variant forest plot.